

BENXI, China

WMO: 543460

Lat: 41.3067N

Lon: 123.7753E

Elev: 607

StdP: 14.38

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.8	(c) -3.8	(d) -17.6	(e) 2.1	(f) 0.8	(g) -13.9	(h) 2.6	(i) 3.0	(j) 13.9	(k) 21.8	(l) 11.9	(m) 22.1	(n) 4.8	(o) 90	(p) 0.434

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.9	(c) 89.4	(d) 72.0	(e) 86.8	(f) 71.4	(g) 84.7	(h) 70.8	(i) 76.9	(j) 84.2	(k) 75.4	(l) 82.4	(m) 74.2	(n) 80.8	(o) 6.1	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.7	133.0	81.5	73.3	126.6	79.5	72.0	120.9	78.7	41.0	84.4	39.5	82.7	38.3	80.9	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)			(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.0	11.8	10.2	DB		-13.0	93.4	5.3	3.6	-16.8	96.0	-19.9	98.1	-22.9	100.1	-26.7	102.7
			WB		-13.5	78.5	5.2	1.8	-17.2	79.7	-20.2	80.8	-23.1	81.8	-26.9	83.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.6	12.9	21.1	35.0	50.5	62.2	71.0	76.4	74.4	64.6	50.5	33.3	18.0	(6)
	DBStd	23.20	8.89	9.48	8.73	8.09	6.25	4.83	3.77	4.65	5.81	8.15	10.31	9.47	(7)
	HDD50	4115	1149	808	468	92	2	0	0	0	1	95	508	992	(8)
	HDD65	7283	1614	1228	929	436	128	8	0	2	77	453	951	1457	(9)
	CDD50	3251	0	0	4	108	381	630	817	756	438	110	7	0	(10)
	CDD65	945	0	0	0	2	42	188	352	294	65	2	0	0	(11)
	CDH74	7567	0	0	0	26	444	1501	2890	2278	417	11	0	0	(12)
	CDH80	2090	0	0	0	1	83	457	859	641	48	1	0	0	(13)
	WSAvg	4.8	4.4	4.8	5.5	6.0	5.6	4.8	4.4	4.1	4.4	4.6	4.7	4.4	(14)
	PrecAvg	31.7	0.3	0.5	0.8	1.9	2.6	4.5	8.0	7.2	2.3	2.0	1.2	0.6	(15)
(16) Precipitation	PrecMax	49.3	1.1	1.6	3.1	4.1	6.9	8.0	18.2	17.9	5.4	5.4	3.1	1.4	(16)
	PrecMin	21.3	0.0	0.0	0.0	0.0	0.7	1.5	3.6	3.0	0.2	0.2	0.2	0.0	(17)
	PrecStd	6.3	0.2	0.4	0.6	1.0	1.4	1.7	2.8	3.4	1.1	1.1	0.7	0.3	(18)
	PrecStd	6.3	0.2	0.4	0.6	1.0	1.4	1.7	2.8	3.4	1.1	1.1	0.7	0.3	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.7	48.6	62.1	77.7	86.2	92.7	93.1	91.9	84.3	75.2	62.6	46.2	(19)
		MCWB	34.0	38.6	45.5	55.8	62.5	66.7	74.4	75.0	68.6	60.0	52.2	41.1	(20)
	2%	DB	35.1	43.3	56.7	72.1	82.3	88.4	89.8	87.8	81.0	70.8	58.2	41.1	(21)
		MCWB	29.9	35.0	43.5	53.0	60.6	66.5	74.2	73.9	65.8	57.6	49.1	35.8	(22)
	5%	DB	31.4	40.2	52.7	68.4	79.1	85.1	86.9	85.7	78.6	67.4	54.0	37.0	(23)
		MCWB	26.7	33.4	41.0	51.5	59.5	66.1	73.4	72.8	64.4	55.8	46.2	32.0	(24)
	10%	DB	27.7	37.0	49.0	64.9	75.7	81.9	84.7	83.6	76.1	64.1	50.1	33.4	(25)
		MCWB	23.4	30.7	38.9	49.4	57.8	65.9	72.8	72.2	63.5	54.1	43.1	29.3	(26)
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.0	40.9	48.8	60.0	66.7	72.5	79.8	78.6	72.2	62.8	53.9	42.0	(27)
		MCDB	38.7	46.4	58.6	71.2	79.7	82.1	88.0	86.5	79.4	71.7	60.7	46.3	(28)
	2%	WB	30.2	36.3	45.1	56.0	63.9	70.4	77.5	76.8	69.8	59.9	49.9	36.2	(29)
		MCDB	34.4	42.0	53.5	67.6	76.8	80.0	85.4	83.5	76.8	67.5	57.2	40.6	(30)
	5%	WB	26.8	33.7	42.3	53.5	61.9	69.2	75.9	75.5	67.7	57.7	46.8	32.5	(31)
		MCDB	31.2	39.3	51.2	65.9	74.4	78.9	83.4	82.2	74.5	65.5	53.2	36.3	(32)
	10%	WB	23.5	31.0	39.5	51.1	60.1	68.1	74.6	74.3	65.7	55.5	43.7	29.3	(33)
		MCDB	27.4	36.8	48.2	62.1	72.0	78.0	81.5	80.8	73.2	62.3	49.4	33.4	(34)
(36) Mean Daily Temperature Range	5% DB	MDBR	19.4	18.6	16.7	18.8	19.8	17.3	13.9	15.1	19.3	19.5	16.8	17.3	(35)
		MCDBR	21.6	19.9	22.0	24.8	26.1	24.5	18.7	18.1	22.1	23.3	21.1	18.8	(36)
		MCWBR	17.0	13.6	12.4	11.8	10.5	8.4	6.5	6.4	8.6	11.5	13.8	15.0	(37)
	5% WB	MCDWR	20.4	17.1	18.9	19.6	20.4	16.8	13.9	13.3	17.0	19.2	18.5	17.6	(38)
		MCWBR	16.6	12.5	12.0	10.4	9.6	6.8	5.9	6.1	8.1	10.5	13.9	15.0	(39)
(40) Clear-Sky Solar Irradiance	taub		0.378	0.445	0.505	0.602	0.642	0.716	0.692	0.601	0.530	0.543	0.461	0.396	(40)
	taud		2.000	1.845	1.786	1.649	1.612	1.538	1.626	1.798	1.892	1.777	1.896	1.987	(41)
	Ebn at Noon		229	229	232	219	215	200	203	218	223	196	195	208	(42)
	Edh at Noon		38	52	62	77	82	88	80	65	56	56	43	36	(43)
(44) All-Sky Solar Radiation	RadAvg		752	1014	1356	1586	1798	1618	1452	1460	1383	1032	713	613	(44)
	RadStd		47	70	88	106	142	172	122	131	95	64	75	48	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page